

Project Proposal

1 Project Title

Speech Emotion Recognition from Audio Using Acoustic Features and Machine Learning

2 Student Information

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3 Project Overview

This project aims to develop a speech emotion recognition system that classifies audio recordings into four emotion categories: happy, sad, angry, and neutral. This project will investigate how emotional information can be represented using acoustic features extracted from speech signals.

4 Background and Motivation

Speech contains both linguistic information and emotional information. Emotional states may affect pitch, energy, speaking intensity, and spectral characteristics.

Speech emotion recognition has potential applications in customer-service analysis, human-computer interaction, and assistive analysis of emotional states. However, emotion recognition can be affected by speaker differences, recording conditions, class imbalance, and differences between acted and natural speech.

This project is closely related to acoustic analysis, speech processing, feature extraction, and pattern classification (El Ayadi et al., 2011).

5 Proposed Methodology

- ⑩ Tools and platforms

The project will be implemented in MATLAB using the ELEC5305 Toolbox provided by the teaching staff. MATLAB's Statistics and Machine Learning Toolbox will be used to develop and compare the classification models.

⑩ Signal processing techniques

Each audio recording will be converted to mono and processed using short overlapping frames with a Hamming window. Optional pre-emphasis will be investigated. Frame-level acoustic features will then be extracted, including Mel-frequency cepstral coefficients (MFCCs), fundamental frequency (F0/pitch), short-time energy, zero-crossing rate, harmonic ratio, spectral centroid and spread, spectral entropy, spectral flux, and spectral roll-off. Statistical summaries, such as the mean and standard deviation, will be calculated across the frames to produce a fixed-length feature vector for each recording.

Support Vector Machine (SVM) and Random Forest classifiers will be trained and compared. An MFCC-only model will be used as a baseline, followed by models that combine MFCCs with pitch, energy, harmonic, and spectral features (Ancilin & Milton, 2021). If time permits, a Convolutional Neural Network using Mel-spectrogram inputs will be explored as an optional extension (Satt et al., 2017). Model performance will be evaluated using accuracy, Macro-F1 score, per-class precision and recall, and confusion matrices.

⑩ Data sources

The primary data source will be the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) (Livingstone & Russo, 2018). Four speech emotion classes will be selected: happy, sad, angry, and neutral. The dataset will be organised using the emotion and actor identifiers contained in the audio filenames. A speaker-independent split will be used so that speakers in the test set do not appear in the training set.

6 Expected Outcomes

The project is expected to deliver a complete and reproducible speech emotion recognition workflow. The main deliverables will include:

- ⑩ An audio preprocessing and acoustic feature-extraction pipeline developed in MATLAB
- ⑩ Trained SVM and Random Forest classifiers for recognising happy, sad, angry, and neutral speech
- ⑩ A comparison of MFCC-only features with combined pitch, energy, harmonic, and spectral feature sets
- ⑩ Model performance results reported using accuracy, Macro-F1 score, and per-class precision and recall
- ⑩ Confusion matrices showing the classification performance for each emotion
- ⑩ An analysis of commonly confused emotion classes and possible causes of classification errors

7 Timeline (Weeks 1-13)

Week	Task
1-2	Finalise topic, scope, repository, and proposal
3-5	Literature review and RAVDESS data preparation
6-7	Audio preprocessing and feature extraction
8-9	SVM and Random Forest implementation
10-11	Evaluation, feature comparison, and optimisation
12-13	Final report, results, and GitHub documentation

8 References

1. Livingstone, S. R., & Russo, F. A. (2018). The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English. *PLOS ONE*, 13(5), e0196391. <https://doi.org/10.1371/journal.pone.0196391>
2. El Ayadi, M., Kamel, M. S., & Karray, F. (2011). Survey on speech emotion recognition: Features, classification schemes, and databases. *Pattern Recognition*, 44(3), 572–587. <https://doi.org/10.1016/j.patcog.2010.09.020>
3. Ancilin, J., & Milton, A. (2021). Improved speech emotion recognition with Mel frequency magnitude coefficient. *Applied Acoustics*, 179, 108046. <https://doi.org/10.1016/j.apacoust.2021.108046>
4. Keesing, A., Koh, Y. S., & Witbrock, M. (2021). Acoustic features and neural representations for categorical emotion recognition from speech. In *Proceedings of Interspeech 2021* (pp. 3415–3419). <https://doi.org/10.21437/Interspeech.2021-2217>
5. Satt, A., Rozenberg, S., & Hoory, R. (2017). Efficient emotion recognition from speech using deep learning on spectrograms. In *Proceedings of Interspeech 2017* (pp. 1089–1093). <https://doi.org/10.21437/Interspeech.2017-200>